

DevOps Syllabus — Till Now

Organized revision checklist based on the topics covered in this conversation.

1. Introduction to DevOps

- What is DevOps?
- Need for DevOps
- DevOps culture
- Development + Operations
- Benefits of DevOps
- DevOps lifecycle

2. DevOps Pipeline

- What is a DevOps pipeline?
- Stages of a pipeline
- Build
- Test
- Release
- Deploy
- Monitor

3. DevOps Blueprint

- DevOps workflow/blueprint
- Development → Testing → Deployment → Operations
- Automation in DevOps
- Collaboration between teams

4. CI/CD

- Introduction to CI/CD
- Continuous Integration (CI)
- Continuous Delivery
- Continuous Deployment
- Difference between CI, Continuous Delivery and Continuous Deployment
- Basic CI/CD pipeline

5. Virtualization

- What is virtualization?
- Need for virtualization
- Types of virtualization
- Hypervisor
- Type 1 hypervisor

- Type 2 hypervisor
- Type 1 vs Type 2
- Virtualization and resource sharing
- Multiplexing
- Emulation

6. Virtual Machines

- What is a Virtual Machine?
- Components of a VM
- VM architecture
- VM lifecycle
- Advantages/disadvantages
- VM provisioning
- VM vs physical machine

7. Cloud Computing

- Introduction to Cloud
- Characteristics of cloud
- Cloud deployment models: Public, Private, Hybrid, Community
- Cloud service models: IaaS, PaaS, SaaS

8. Microsoft Azure

- Introduction to Azure
- Azure cloud services
- Azure virtual machines
- Azure infrastructure/resources

9. Infrastructure as Code (IaC)

- What is IaC?
- Why IaC?
- Benefits of IaC
- Infrastructure through code
- Declarative vs Imperative
- Provisioning vs Configuration
- IaC tools

10. Terraform

- What is Terraform?
- Terraform and IaC
- Declarative infrastructure
- Terraform configuration
- Terraform provisioning

- Basic Terraform code
- Terraform workflow
- Terraform state — if included in your class
- Terraform vs Ansible

11. Ansible

- What is Ansible?
- Configuration management
- Automation
- Ansible playbooks
- YAML
- Provisioning vs configuration
- Basic Ansible configuration code
- Terraform vs Ansible

12. Provisioning & Configuration

- What is provisioning?
- What is configuration?
- Difference between provisioning and configuration
- Provisioning tools
- Configuration-management tools
- Basic provisioning code
- Basic configuration code

13. Docker

- What is Docker?
- Why Docker?
- Docker architecture
- Docker Engine
- Docker images
- Docker containers
- Docker Hub
- Docker commands
- Docker image execution code
- Creating/running containers

14. Containerization

- What is containerization?
- How containerization works
- Benefits of containerization
- Containers vs Virtual Machines

- Images vs containers
- Container lifecycle

15. Containers vs Virtual Machines

- VM: complete guest OS, heavier, slower startup, more resource usage, hypervisor-based
- Container: shares host OS kernel, lightweight, faster startup, less resource usage, container-engine based, process-level isolation

16. RAID

- What is RAID?
- Need for RAID
- RAID levels
- RAID 0
- RAID 1
- RAID 5
- RAID 6
- RAID 10
- Striping
- Mirroring
- Parity
- Fault tolerance
- Performance vs redundancy

Overall Concept Flow

DEVOPS	→	DevOps Pipeline	→	CI/CD
Infrastructure	→	Virtualization	→	Virtual Machines
Cloud	→	Azure	→	IaC
IaC	→	Terraform	→	Provision
IaC	→	Ansible	→	Configure
Docker	→	Containerization	→	Containers vs VMs
RAID		Storage / Fault Tolerance		

Key memory points

- Terraform → primarily declarative; used mainly to provision infrastructure.
- Ansible → used mainly for configuration and automation.
- Docker → packages and runs applications in containers.
- VM → virtual computer with a guest operating system.
- Container → isolated application environment that shares the host OS kernel.
- Imperative → HOW to do something; Declarative → WHAT desired state is required.
- Provisioning → create/provide resources; Configuration → set up existing resources.